Week 10

Perception

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Today

0:00 – 0:50	Lecture
1:00 – 2:00	**Track milestone demos** — 8 min per team
2:00 – 2:30	Track work
2:30 – 3:00	Standup — **Merge 1 next week**

1. What the sensors measure

2D LiDAR

Range to the nearest surface, along a set of rays,
in a single horizontal plane. That is all.

Samples	720 (0.5° resolution)
Field of view	360°
Range	12 m
Rate	10 Hz

What a horizontal-plane scanner cannot see

- **Below the plane** — a kerb, a step, a hole
- **Above it** — a table edge, a low shelf, an open drawer
- **Transparent or mirrored** — glass doors read as open space

And on this vehicle: the LiDAR **faces forward.**
There is no rear sensing at all.

The map does not distinguish nothing there from nothing visible.

SLAM and Nav2 both treat "no return" as free space.

Camera

Colour and texture, projected onto a plane. **No depth.**

Resolution	640 × 480
Horizontal FOV	1.396 rad (~80°)
Rate	30 Hz

Rolling shutter

The sensor exposes the image **row by row**, not all at once.

Move during the scan and the image **shears** —
vertical lines lean, and the lean scales with speed.

The ADR chooses a **$30 rolling-shutter USB camera** for semester 1.

Why that is a good decision, not a cheap one

At ≤ walking pace, on a vehicle whose limiting factor is **feedback quality** rather than image quality, shear is small.

And the ADR names what would flip it:

Phase 2 is behavior cloning. The camera stops being something you look at and becomes the **input to a learned policy.** A speed-dependent geometric distortion in the training data is then a systematic bias the network will happily learn — and it will not show up in the loss curve.

2. Calibration

Intrinsics — the camera's own geometry

Focal length, principal point, lens distortion.
The mapping from **3D rays to pixels.**

Without it you cannot turn a pixel into a direction,
so you cannot relate the image to anything else in the world.

Procedure: a checkerboard, many views, varied distance and angle.

The twin cannot teach you this

In simulation the camera is a **perfect pinhole** and its `camera_info` is exact.

So intrinsic calibration is free there — not because it is easy, but because **it does not exist as a problem.**

Worth noticing rather than enjoying.

Extrinsics — where the sensor is

Intrinsics are about the sensor. Extrinsics are about the **vehicle**.

In simulation you typed the numbers into the URDF and they became true.

On hardware it runs the other way: the sensor is bolted somewhere,
and you must **measure** where.

The staged procedure

Stage	
Coarse	Ruler from the `base_link` origin. Good to ~1 cm
LiDAR refinement	Drive parallel to a flat wall; the scan must be parallel too
Camera→LiDAR	Project scan points into the image; they must land on their objects

Lab 5 already showed you the cost

A **5 cm** error. Nothing errors. LiDAR publishes, TF is complete, SLAM runs — and walls observed from different headings stop landing on top of each other.

A **constant** mounting offset makes a **heading-dependent** map error, because that offset points in a different world direction as the car rotates.

Which is why the gate is stated as **"the same wall within 10 cm"** rather than as something about the sensor.

Time sync is not a formality

If the LiDAR's timestamps are offset from the vehicle's, TF happily interpolates a pose for the **wrong instant.**

Every scan is placed slightly wrong — **in a way that scales with speed.**

So it looks like a mechanical problem, not a clock problem.

(This is also why ADR-SW1 insists on one transport and one clock.)

Track milestone demos — 8 min each

Show one thing working end to end, **with a number attached.**

Track	A good answer
Electrical	Steering holds an angle — here is the steady-state error and loop rate
Chassis	Measured wheelbase, track, wheel radius — and how each was measured
Simulation	`mitt_bench` reports odometry drift over 20 m — here is the number
Software	`mitt_hardware` loads, activates, round-trips a recorded bag

Merge 1 is next week

Electrical + Chassis: the vehicle steers to a commanded angle, wheels off the ground. Closes bring-up **Stage 1**.

Anything that will not be ready must be raised **today.**

Chassis: wheel radius is measured **loaded, on the floor.**
Hollow plastic wheels deflect under 6 kg of payload.